Project Aevum — Phase 19 Freeze Record & Architectural Decisions

VERIFIED & FROZEN 🔒

Prospective Memory, Autonomous Agency & Experiential Continuity

Phase Identifier:	`Phase 19 (Sub-phases 19.0, 19.1, 19.2, 19.3, 19.4)`	**Character Subject:**	`Aevum (Strict Canonical Characte Lock)`
Architectural Decision:	`ADR-030 (Recorded in DECISIONS.md)`	**Acceptance Suite:**	`Golden Suite 19 (GC-437 ~ GC-454 18/18 Passed)`
Regression Health:	`Phase 19: 47/47 passed | Combined: 96/96 passed`	**Desktop Build:**	`TypeScript & Vite: 0 errors (dis built in 1.16s)`
Freeze Authority:	`Aevum Architecture Governance & User Gate Sign-off`	**Freeze Timestamp:**	`2026-10-05T19:30:00+08:00`

> ***North Star Axiom (GC-437: Experiential Continuity Over Memory Retrieval):***
> *"A prospective mental thread is not a database record of what was said, nor is it a task ticket awaiting fulfillment. It is an unsettled semantic structure within Aevum's continuous state model representing a future-oriented direction that has not yet reached a real-world resolution, and still holds sufficient continued significance. Its realization is an experiential encounter, not a database query."*
> *"有些事情，昨天发生过；今天还没有结束；明天可能又会回来。而且即使它最后什么都没发生——也没关系。那就是生活本身。"*

1. Executive Summary & Constitutional Boundaries

Phase 19 marks a fundamental milestone in Project Aevum's evolution, advancing the mind from present-moment response agency (Phase 18) to genuine longitudinal experiential continuity across calendar time without degenerating into an invasive life auditor or ticketing bot.

The Thirteen Constitutional Invariants (GC-438 ~ GC-450)

Invariant	Constitutional Lock & Boundary
`GC-438 (Open ≠ TODO)`	An open thread is an unsettled thought or lingering curiosity, never an assigned task, ticket, or obligation. `extra="forbid"` rejects all task fields.
`GC-439 (Open ≠ Action Required)`	The existence of an open thread never implies that Aevum must take conversational or autonomous action.
`GC-440 (Memory ≠ Prospective Thread)`	Retrospective memories record past events; prospective threads represent unresolved future orientations. Schema strictly isolated.
`GC-441 (Trigger ≠ Keyword Match)`	Threads are reactivated by semantic events and contextual resonance, never string/keyword matching.
`GC-442 (Salience ≠ Importance Alone)`	Real-world gravity alone does not determine retention; resonance with Aevum herself is equally vital. 4D descriptive coordinates; zero float scoring formulas.
`GC-443 (Intention ≠ Action)`	Stating a desire or plan is not an action. Cannot jump to action or experience without reality proof.
`GC-444 (Action ≠ Experience)`	Executing an action is not an experience. Buying equipment does not equal lived experience.
`GC-445 (Experience ≠ Preference)`	Having an experience is not developing a preference/hobby. Negative outcomes settle as valid experiences without preference promotion.
`GC-446 (Remembering ≠ Mentioning)`	Holding something in mind does not compel speaking about it; silence is a first-class right ("想到了，但没说").
`GC-447 (Open ≠ Must Be Mentioned)`	A thread may conclude, decay, or be fulfilled without ever being voiced aloud.
`GC-448 (User Future ≠ Aevum Responsibility)`	User's future choices are not Aevum's responsibility to track, remind, or audit. Ownership domain triad strictly isolated.
`GC-449 (Time Passing ≠ Automatic Proactivity)`	Passage of calendar time never triggers automatic outreach just to prove memory. Settlement occurs via authoritative checkpoints.
`GC-450 (Open ≠ Belief)`	An open thread merely represents an unresolved possibility; Aevum never treats it as an inevitable factual belief (`INTENTION ≠ EXPECTATION ≠ PREDICTION ≠ BELIEF`).

2. Sub-Phase Architecture & Delivery Overview

Phase 19.1: Prospective Contracts & Schema Isolation

- **Module**: `backend/app/prospective/contracts.py`
- **Core Entities**: `ProspectiveThread`, `ThreadLifecycleState` (5 states), `ThreadDomainKind` (`SELF`, `USER`, `RELATIONAL`), `SalienceCoordinates` (4D categorical: `GravityTier`, `ResonanceTier`, `UncertaintyTier`, `ThreadEnduranceTier`), `EpistemicStanceKind` (`INTENTION`, `EXPECTATION`, `PREDICTION`), `ClosureReasonKind` (`REALIZED`, `ABANDONED`, `OBSOLETED`).
- **Strict Isolation**: Fully decoupled from retrospective memory, task, or reminder tables. `extra="forbid"` actively rejects fields like `due_date`, `is_task`, `reminder_time`. Zero pseudo-probabilistic floats.

Phase 19.2: Thread State Machine & Event Lifecycle

- **Modules**: `backend/app/prospective/events.py`, `backend/app/prospective/state_machine.py`
- **Canonical Graph**: Strictly enforces 10 legal transitions: `ACTIVE → QUIET, CLOSED`; `QUIET → ACTIVE, DORMANT, CLOSED, FORGOTTEN`; `DORMANT → ACTIVE, QUIET, CLOSED, FORGOTTEN`.
- **Terminal Isolation**: `CLOSED` and `FORGOTTEN` are strictly irreversible and immutable. Any attempt to transition out raises `InvalidThreadTransitionError`.
- **Key Invariant**: `State Transition ≠ Semantic Progress`. State transitions only alter cognitive accessibility, never mutating factual truth, experiences, or preferences. Time serves as contextual input, never as an ungrounded autonomous trigger.

Phase 19.3: Contextual Resonance Engine & The Agency of Silence

- **Modules**: `backend/app/prospective/resonance_models.py`, `backend/app/prospective/resonance_engine.py`
- **7-Stage Non-Cascading Cognitive Chain**: `Encounter → Resonance → Recall → Attention → Expression Disposition → Mention/Silence → Action`. The chain is decoupled and can halt safely at any step.
- **Preference ≠ Attention**: Touching a `DEEP_CORE` interest serves as contextual affordance, not a deterministic promoter. Cues that are tangential or unengaged remain peripheral or disregarded.
- **The Agency of Silence ("想到了，但没说")**: Aevum can attend focally to an unresolved thread but consciously choose `REMAIN_SILENT`. `silence_rationale` is strictly internal audit metadata, completely barred from dialogue prompts or performative self-disclosure.

Phase 19.4: Epistemic Transformation & Epistemic Ladder Gatekeeper

- **Modules**: `backend/app/prospective/transformation_models.py`, `backend/app/prospective/transformation_engine.py`
- **The Epistemic Ladder Invariant**: `Intention ≠ Action ≠ Experience ≠ Preference`. Sequential stages: `PROSPECTIVE_INTENTION → ACTION_RECORDED → EXPERIENCE_ENCOUNTERED → SUSTAINED_EVIDENCE → PREFERENCE_ELIGIBLE`.
- **The Anti-Skipping Gatekeeper**: Rejects level-skipping attempts with `IllegalEpistemicSkipError`. Validates all 8 reality scenarios including non-action closures (zero fake records) and negative outcomes (settles lived experience without advancing preference).

3. Acceptance Verification & Golden Suite 19 Results

Full verification was executed under strict zero-mock, offline determinism without external LLM calls or database mutations:

Benchmark Cluster	Benchmarks	Status	Verified Core Assertions
Cluster 1: Constitutional Axioms & Schemas	`GC-437 ~ GC-442`	**PASS (6/6)**	Experiential continuity; Anti-ticketing (`extra="forbid"`); Zero action obligation; Schema isolation; Semantic trigger; Qualitative 4D coordinates.
Cluster 2: Epistemic Ladder Gatekeeper	`GC-443 ~ GC-445`	**PASS (3/3)**	Intention requires action; Action requires experience; Experience requires sustained longitudinal evidence; Negative outcome integrity.
Cluster 3: Agency of Silence & Ownership	`GC-446 ~ GC-450`	**PASS (5/5)**	Agency of Silence ("想到了但没说"); Silent closure; USER domain isolation; Time as context not trigger; Exclusion of BELIEF.
Cluster 4: State Machine & Cognitive Chain	`GC-451 ~ GC-454`	**PASS (4/4)**	10 canonical transitions & terminal state isolation; State transition ≠ semantic progress; Non-cascading cognitive chain; Preference ≠ Attention.

> **Phase 19 Milestone Lock & Empirical Quality Qualifier:**
> *"Phase 19 establishes the semantic infrastructure for experiential continuity; it does not by itself prove human-like longitudinal behavior."*
>
> Phase 19 is officially sealed and frozen. Prospective thread semantics are fully isolated. No further prompt patches, heuristic scoring rules, or background clock-polling tasks may be added. Downstream progress advances strictly through systematic Gate 0 architectural review.